SOCIOL 723: Social Statistics II

Week 11, Synthetic Control and Its Extensions

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Today

The Synthetic Control Estimator

Extensions

When There Is Only One Treated Unit

Difference-in-differences needs a control *group*. What if the treated unit is California, or East Germany, or a single firm?

- Comparative case studies have always done this informally: "compare California to the rest of the country." The choice of comparison was made by the author, and the reader had to trust it.
- Synthetic control makes that choice a data-driven, transparent, pre-registered procedure: construct a weighted average of untreated units that reproduces the treated unit's pre-treatment trajectory.
- "Arguably the most important innovation in the policy evaluation literature in the last 15 years" (Athey and Imbens 2017).

The identifying idea is not new. What is new is that the weights are chosen by an algorithm, reported in a table, and checked against placebo units.

Synthetic Control

Setup

Unit 1 is treated from period $T_0 + 1$ onward. Units $2, \ldots, J+1$, the donor pool, are never treated. The estimand is

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0), \qquad t > T_0.$$

We observe $Y_{1t}(1)$; the whole problem is imputing $Y_{1t}(0)$.

The synthetic control

Choose weights $w_2, \ldots, w_{J+1}$ with $w_j \geq 0$ and $\sum_j w_j = 1$ so that the weighted donors match the treated unit *before* treatment:

$$\sum_{j \geq 2} w_j Y_{jt} \approx Y_{1t} \;\text{ for } t \leq T_0, \qquad \sum_{j \geq 2} w_j X_j \approx X_1,$$

then estimate $\hat{\tau}_{1t} = Y_{1t} - \sum_{j \geq 2} w_j Y_{jt}$ for $t > T_0$.

The Optimization Problem, Explicitly

Let $\boldsymbol{X}_1$ be a $k \times 1$ vector of pre-treatment predictors for the treated unit and $\boldsymbol{X}_0$ the $k \times J$ matrix for the donors. Choose weights

$$\boldsymbol{W}^*(\boldsymbol{V}) = \underset{\boldsymbol{W} \in \mathcal{W}}{\arg\min}\ (\boldsymbol{X}_1 - \boldsymbol{X}_0\boldsymbol{W})'\, \boldsymbol{V}\, (\boldsymbol{X}_1 - \boldsymbol{X}_0\boldsymbol{W}), \qquad \mathcal{W} = \left\{\boldsymbol{W} \geq 0,\ \textstyle\sum_j w_j = 1\right\}.$$

- $\boldsymbol{V}$ is a diagonal matrix of predictor importance weights. It is itself chosen, usually to minimize pre-treatment prediction error of the *outcome*: $\boldsymbol{V}^* = \underset{\boldsymbol{V}}{\arg\min} \sum_{t \leq T_0} \left(Y_{1t} - \sum_j w_j^*(\boldsymbol{V}) Y_{jt}\right)^2$.
- So there are two nested optimizations, and the inner one is a constrained quadratic program. The constraint set $\mathcal{W}$ is a simplex.
- This nesting is where researcher degrees of freedom hide. Which predictors enter $\boldsymbol{X}$, and how many pre-periods are used, can move the answer. Pre-register the choice or report the full space.

The Factor-Model Justification (cont.)

Matching California's *past* is only useful if it tells us something about California's missing *future*. Why should it? The factor model is the answer: it states the conditions under which a good pre-treatment fit licenses that extrapolation.

Suppose untreated outcomes follow a linear factor model

$$Y_{jt}(0) = \delta_t + \boldsymbol{\theta}_t' \boldsymbol{Z}_j + \boldsymbol{\lambda}_t' \boldsymbol{\mu}_j + \epsilon_{jt},$$

with $\boldsymbol{\lambda}_t$ common (unobserved) factors and $\boldsymbol{\mu}_j$ unit-specific loadings.

- **DID** allows arbitrary common time shocks δ_t, but requires that the remaining term $\boldsymbol{\lambda}_t' \boldsymbol{\mu}_j$ not generate *differential* trends between treated and control units. Synthetic control weakens that by matching on the loadings instead.

The Factor-Model Justification (cont.)

- Abadie, Diamond, and Hainmueller (2010) show that if a convex combination of donors reproduces the treated unit's outcomes over a *long* pre-period *and* its observed predictors, then under regularity conditions, enough pre-periods relative to the noise, and a factor structure of bounded rank, the weighted donors also approximately match the unobserved loadings $\boldsymbol{\mu}_1$, and a bound on the post-treatment bias shrinks with the number of pre-periods.
- Length of the pre-period is doing the identification work. A two-period match proves nothing; a twenty-period match is informative. This is the single most important design consideration.

Why the Constraints Matter

The restrictions $w_j \geq 0$ and $\sum_j w_j = 1$ look like technicalities. They are the heart of the method.

- **No extrapolation**: the synthetic unit is a *convex combination* of real units, so it lies inside the donor pool's support. Unconstrained regression would happily assign weights of $+3$ and -2 and predict outside anything ever observed.
- **Sparsity**: the constraints usually produce weights on a handful of donors and zero on the rest. You can print the weight table and let readers judge whether the comparison is sensible.
- **Transparency**: the counterfactual is a named, weighted set of real places, not a regression coefficient.
- If no convex combination matches the treated unit, if California is richer than every donor, the method tells you, through poor pre-treatment fit. That is a feature.

Relation to Difference-in-Differences

DID is synthetic control with the weights fixed at $1/J$ (or population shares), and with an added intercept shift.

- DID assumes parallel trends between the treated unit and a *pre-specified* average of controls.
- Synthetic control *searches* for a weighting under which pre-treatment trajectories coincide, and then assumes that coincidence continues.
- The formal justification is a linear factor model $Y_{jt}(0) = \delta_t + \boldsymbol{\theta}_t' Z_j + \boldsymbol{\lambda}_t' \boldsymbol{\mu}_j + \epsilon_{jt}$: if the synthetic control matches the treated unit on a long pre-period; it must also match on the unobserved factor loadings $\boldsymbol{\mu}_j$ (Abadie, Diamond, and Hainmueller 2010).
- Length of the pre-period is doing the identification work. A two-period pre-window proves nothing; a twenty-period match is informative.

Inference With One Treated Unit (cont.)

Conventional standard errors are meaningless here: $n_{\text{treated}} = 1$, and the usual asymptotics run in the number of treated units. Instead, permutation inference.

Placebo (in-space) tests

Pretend each donor was the treated unit and estimate its "effect." If the real treated unit's post-treatment gap is extreme relative to the placebo distribution, that is evidence.

For each unit j compute

$$\text{RMSPE}_j^{\text{pre}} = \sqrt{\tfrac{1}{T_0} \sum_{t \le T_0} \left(Y_{jt} - \hat{Y}_{jt}^N\right)^2}, \qquad \text{RMSPE}_j^{\text{post}} = \sqrt{\tfrac{1}{T - T_0} \sum_{t > T_0} \left(Y_{jt} - \hat{Y}_{jt}^N\right)^2},$$

and rank units by the ratio $\text{RMSPE}_j^{\text{post}} / \text{RMSPE}_j^{\text{pre}}$.

Inference With One Treated Unit (cont.)

- The ratio, rather than the raw post-treatment gap, is what makes placebos comparable: a donor the method could not fit will show a large gap for reasons unrelated to treatment. **Discard poorly fitting placebos.**
- The rank of the treated unit among the $J+1$ units gives a permutation-style p-value: with 38 donors the smallest attainable value is $1/39 \approx 0.026$. Call it a placebo rank, not an exact p-value: exactness would require treatment to have been randomly assigned among these units under a sharp null, which it was not.
- **In-time placebo**: re-run pretending treatment began earlier. A gap should not appear before the real intervention.
- **Leave-one-donor-out**: drop each positively weighted donor in turn. If the result rests on one unit, say so.

The Canonical Application: California Proposition 99 (cont.)

Abadie, Diamond, and Hainmueller (2010) study California's 1988 tobacco control programme.

- **Outcome**: per-capita cigarette sales, 1970–2000. **Donor pool**: the 38 states that never adopted comparable programmes over the period, states with their own large tax increases or programmes are excluded *a priori*, on substantive grounds.
- **Synthetic California** is a weighted average of Colorado, Connecticut, Montana, Nevada, and Utah. The weight table is printed in the paper, and readers can judge whether that is a sensible California.
- Pre-1988 trajectories are nearly indistinguishable; afterwards they diverge sharply, implying roughly 26 fewer packs per capita per year by 2000.

The Canonical Application: California Proposition 99 (cont.)

- In the placebo distribution, California's post/pre RMSPE ratio is the largest of all 39 units, giving $p = 1/39$.

Notice how much of the credibility comes from things a reader can check by eye: the weight table, the pre-treatment overlap, and the placebo plot.

Diagnostics and Failure Modes

- **Poor pre-treatment fit** invalidates the design. Report the pre-treatment RMSPE and plot the two trajectories. If they do not overlap before treatment, stop.
- **Interpolation bias**: donors very unlike the treated unit can be combined to match on average while differing on everything that matters. Restrict the donor pool on substantive grounds *before* estimating.
- **Overfitting the pre-period**: matching noise rather than structure. A long pre-period and few donors is the safe configuration.
- **Contamination**: donors affected by the treatment (spillovers) or by their own interventions must be excluded.
- **Specification search**: the choice of predictors and pre-periods can move results. Pre-register it, or report the full space.

Extensions

When the Convex Hull Is Empty

- **Augmented synthetic control** (Ben-Michael, Feller, and Rothstein 2021): when no convex combination fits, allow a *bias correction*, fit an outcome model to the residual imbalance, exactly the augmentation logic of AIPW in Week 6. Relaxes the no-extrapolation constraint by a controlled amount.
- **Ridge-augmented** versions penalize how far the weights stray from uniform, trading a little extrapolation for much better fit.
- **Generalized synthetic control** (Xu 2017): estimate an interactive fixed effects model $Y_{it}(0) = \boldsymbol{\lambda}_t' \boldsymbol{\mu}_i + \dots$ on the untreated units and use it to impute $Y_{it}(0)$ for the treated. Handles *multiple* treated units and staggered timing, and gives conditional standard errors via a parametric bootstrap.
- **Synthetic DID** (Arkhangelsky et al. 2021): unit weights *and* time weights, with a DID-style intercept. Doubly robust in spirit and often the best default now.

Choosing Among Them

Situation	Method	R
One treated unit, good pre-fit	classic synthetic control	`Synth`, `tidysynth`
One treated unit, poor pre-fit	augmented / ridge SC	`augsynth`
Several treated units, staggered	generalized SC	`gsynth`
Many units, want DID robustness	synthetic DID	`synthdid`
Many units, staggered, covariates	Callaway–Sant'Anna (Wk 10)	`did`

- These are not rivals so much as points on a spectrum between "trust the weights" and "trust the outcome model." The augmented versions deliberately use both, the same insurance logic as AIPW.
- Report the classic estimator alongside whichever you prefer, so readers can see how much the extrapolation is doing.

Reporting a Synthetic Control Study

1. The donor pool, and why each unit is in it or excluded.
2. The predictors used for matching and the pre-treatment window.
3. **The weight table.** Which units, with what weights.
4. A balance table: treated versus synthetic on every predictor.
5. The trajectory plot, treated versus synthetic, with the intervention marked.
6. The gap plot, with placebo gaps overlaid.
7. Pre- and post-treatment RMSPE, their ratio, and the permutation p-value.
8. In-time placebo and leave-one-donor-out robustness.

The trajectory plot is the result. If the pre-treatment lines do not sit on top of one another, no test rescues the design.

Looking Ahead

- Synthetic control completes the design-based half of the course. Weeks 5–11 have all been variations on one question: *where does the counterfactual come from?*
- Notice the recurring structure: choose weights on comparison units (Week 6), choose an exogenous slice of variation (Weeks 7, 9), choose a differencing scheme (Week 10), choose a weighted donor pool (Week 11).
- **Weeks 12–13** change the question again: what happens when prediction rather than identification is the goal, and what, if anything, flexible prediction can contribute back to causal work.

Final projects

Article sign-up and confirmation of replication data availability due **November 2**.

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